

Stelios Zacharioudakis

MACHINE LEARNING RESEARCH & ENGINEERING

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SUMMARY

Machine learning engineer with a BSc in Computer Science and pro bono experience delivering a member-services platform. Build PyTorch research tools, reproducible LLM evaluations and deployed data applications; current independent research explores latent world models.

WORK EXPERIENCE

Paphos Medical Association

Jun 2022 - Jul 2026

Head Engineer (Pro Bono)

Paphos, Cyprus

- Delivered and donated AsklepiosMed, giving the Paphos Medical Association its own web platform for member services and administration without charging an engineering fee.
- Enabled web-based member and administration tasks by integrating React interfaces, a Node.js/Express API and PostgreSQL, and deploying the complete application on the association's Linux VPS.

SELECTED SOFTWARE PROJECTS

ForgeRL / ForgeBench [ForgeRL](#) / [ForgeBench](#)

- Built isolated coding-agent evaluation and durable budget controls. A frozen prospective pilot completed 162 episodes across 27 miniature tasks and six policies, with full traces and a separate source-derived track. One seed and six test families limit generalization; model weights remain unchanged.

MTA Scan [MTA Scan](#)

- Built a Mapbox transit workspace with autonomous collection and 15/30-minute evaluation of feed-predicted arrival-spacing proxies. Uses chronological holdouts and baselines; early feasibility only, no validated incident claim.

Inference Systems Lab [Inference Systems Lab](#)

- Ran 1,536 measured requests on an RTX 4090 with pinned vLLM/Qwen versions, concurrency sweeps and direct GPU telemetry. Recorded prefix-cache comparisons and output mismatches; no lossless speedup claim.

RESEARCH

Score-based MRI reconstruction

BSc thesis and reproducible evidence companion

Reimplemented a published MRI reconstruction method in PyTorch with Flax checkpoint conversion. The thesis reports PSNR agreement within 0.15 dB on 256 BraTS slices at 4x, 8x and 24x undersampling. Released CPU evidence and ensemble-diagnostic tools; original prior credited, retrospective research.

Latent world-model learning and evaluation

Independent study · ongoing and unpublished

Investigate imagination horizons, critic stability and policy selection with PyTorch latent world models. Design controlled ablations and paired simulator evaluations, retaining task-dependent outcomes and failed hypotheses.

EDUCATION

Bachelor of Science, Computer Science

Jun 2026

National and Kapodistrian University of Athens

- Department of Informatics and Telecommunications

TECHNICAL SKILLS

- **ML:** Python, PyTorch, NumPy, SciPy, scikit-learn, generative models, model-based RL, uncertainty analysis and experimental evaluation
- **Systems:** FastAPI, Node.js/Express, React, JavaScript, SQL/PostgreSQL, Redis, Linux, Nginx, Docker, Git, automated testing